

Memory & Performance Cheat Sheet

Is the box short on CPU, memory or disk? Read free, vmstat, iostat and the load average to find out.

Reading free -h

	total	used	free	shared	buff/cache	available	
Mem:	15Gi	4.6Gi	612Mi	284Mi	9.9Gi	9.8Gi	← 9.8Gi: plenty
Swap:	4.0Gi	256Mi	3.8Gi				

Column	Means	Column	Means
used	Taken by programs	buff/cache	Disk cache. Linux hands it back when needed
free	Totally unused. Low is normal	available	What new programs can get. Watch this one
shared	tmpfs and shared memory (/dev/shm)	Swap used	Some is fine; growing fast = RAM is short

Reading vmstat 1 (one line per second; ignore the first, it's since boot)

procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----																		
r	b	swpd	free	buff	cache	si	so	bi	bo	in	cs	us	sy	id	wa	st	gu	
1	0	262144	626688	210432	9961472	0	0	3	41	512	880	6	2	91	1	0	0	← healthy
9	3	262144	98304	21040	812646	420	980	5120	18200	4810	9920	38	12	12	38	0	0	← trouble

Column	Means (trouble sign in bold)	Column	Means (trouble sign in bold)
r	Wants a CPU. Stays above nproc	us sy id	CPU %: programs / kernel / idle
b	Blocked, waiting on disk or network	wa	CPU idle, waiting on disk. Stays above ~10-20
si / so	Swap in / out. Non-zero for a while	st / gu	Stolen by the hypervisor (VMs) / running guests
swpd...cache	Memory in KiB (-S M for MB)	bi bo in cs	Blocks read, written · interrupts, context switches

Reading iostat -xz 1 (sysstat package; some columns left out)

Device	r/s	rkB/s	r_await	w/s	wkB/s	w_await	aqu-sz	%util	
nvme0n1	210.0	26880.0	0.31	480.0	61440.0	0.85	0.47	38.2	← fast, idle-ish
sda	12.0	384.0	18.40	140.0	8960.0	95.20	13.60	99.8	← saturated

Column	Means	Column	Means
r/s w/s	Reads / writes per second	aqu-sz	Average queue length. Well above 1 = backlog
r_await w_await	ms per request. High = slow disk	%util	Busy %: 100 = maxed on HDD, not on SSD/NVMe

Load average and the OOM killer

\$ uptime
14:02:11 up 12 days, 3:40, 2 users, load average: 7.85, 6.10, 3.02 ← 1, 5, 15 min · compare to nproc
\$ sudo dmesg -T grep -i 'out of memory' ← or: journalctl -k grep -i oom
[Sat Oct 3 02:14:07 2026] Out of memory: Killed process 1234 (java) total-vm:8123456kB, ...

Load counts processes running **or** stuck waiting on disk. On 8 CPUs, 7.85 is busy but OK; 16 is overloaded.

Quick triage

Is it...	Check with	Trouble looks like
CPU?	uptime, top (press 1), mpstat -P ALL 1	Load above nproc , id near 0, vmstat r high
Memory?	free -h, vmstat 1, dmesg -T grep -i oom	available low, si/so non-zero, OOM kills
Disk?	iostat -xz 1, df -h, df -i	High wa and await , a full filesystem
Network?	ip -s link, ss -s, sar -n DEV 1	Errors/drops climbing, link at its speed limit
Swap?	swapon --show, sysctl vm.swappiness	Filling up. Prefer RAM: vm.swappiness=10
Which PID?	top (P / M to sort), pidstat 1, iotop	One PID hogging CPU, RAM or disk

Remember: watch **available**, not **free**. Linux fills spare RAM with disk cache on purpose, so **free** is always low on a healthy box. Worry when **available** is low **and** **si/so** keep ticking.